

Strengths

CSTM enables mixed-camera training

First method to train metric depth on 10K+ camera models without divergence

Zero-shot metric depth

Beats fine-tuned methods on NYU (0.047 AbsRel) and KITTI (0.042) without seeing those datasets

Joint depth + normal prediction

Shared ConvGRU refinement produces both outputs from a single model

L_{d-n} consistency loss

Bridges the 475:1 data gap between depth and normal labels without needing GT normals

Architecture-agnostic CSTM

Pluggable module that works with any backbone (ConvNeXt, ViT)

Scale: 16M images, 18 datasets

Largest metric depth training set at time of publication